Data Cleaning

Cleaning the SVPrecinct data

From the website : - CNG stands for "Congressionnal" - There can be several candidates within each district. This is what we care about. - CDDIST is the congressional district number - SVPREC is the precinct identifier

Simplifying the dataset

We can first select the columns we care about.

```
svprecinct <- read.csv("/Users/sachabondeville/Documents/stat133/Projects/
Project-5/gerrymandering-sachabdvl/data/g24_sov_by_g24_svprec.csv")

sv_precinct <- svprecinct |>
    select(COUNTY, SVPREC, CDDIST, TOTVOTE, CNGDEM01, CNGDEM02, CNGREP01,
CNGREP02)
```

We can also rename columns:

```
sv_precinct <- sv_precinct |>
    rename(DEM01= CNGDEM01,
    DEM02 = CNGDEM02,
    REP01 = CNGREP01,
    REP02 = CNGREP02,
    DISTRICT = CDDIST
          )
```

For our analysis, we particularly want to focus on precinct with sometype of democrat / republican competition. So why do we care about about the DEM02 and REP02 columns? Because there are the best proxy to filter out the cases when there are competition. Notice when you have two democratic candidates, then it means you do not have a republican candidate!

Pattern recognition within column.

Notice CNGs columns are strings. We need to clean that!

```
sv_precinct <- sv_precinct  |>
    mutate(DEM01 = as.double(str_extract_all(DEM01, "\\d+")),
           DEM02 = as.double(str_extract_all(DEM02, "\\d+")),
           REP01 = as.double(str_extract_all(REP01, "\\d+")),
           REP02 = as.double(str_extract_all(REP02, "\\d+")))
```

Getting rid of irrelevant columns.

Notice there are some precinct with no competition (only voting for democratic candidates or republican candidates). For the sake of our analysis, we will then choose to disregard them for some metrics. To simplify our analysis, we will also want a total vote column that only consider democrat and republican votes.

```
sv_precinct <- sv_precinct |>
    mutate(TOTVOTE = REP01+DEM01+REP02+DEM02)
```

District name analysis

```
levels(factor(sv_precinct$DISTRICT))
```

```
 [1] "0"  "1"  "2"  "3"  "4"  "5"  "6"  "7"  "8"  "9"  "10" "11" "12" "13"
"14"
[16] "15" "16" "17" "18" "19" "20" "21" "22" "23" "24" "25" "26" "27" "28"
"29"
[31] "30" "31" "32" "33" "34" "35" "36" "37" "38" "39" "40" "41" "42" "43"
"44"
[46] "45" "46" "47" "48" "49" "50" "51" "52"
```

The district 0 does not exist ! We choose to disregard it. It combines some summary statistics we do not want.

```
sv_precinct <- sv_precinct|>
    filter(DISTRICT != 0)
```

Saving the cleaned dataset

```
write.csv(sv_precinct, "/Users/sachabondeville/Documents/stat133/Projects/
Project-5/gerrymandering-sachabdvl/data/cleaned_svprecinct.csv", row.names =
FALSE)
```

Cleaning the SRPrecinct file

Following the exact same methodology than above

```
sr_precinct <- read.csv("/Users/sachabondeville/Documents/stat133/Projects/
Project-5/gerrymandering-sachabdvl/data/state_g24_sov_data_by_g24_srprec.csv")

sr_precinct <- sr_precinct |>
    select(COUNTY, SRPREC_KEY, CDDIST, TOTVOTE, CNGDEM01, CNGDEM02, CNGREP01,
CNGREP02)|>
    rename(DEM01= CNGDEM01,
```

```
    DEM02 = CNGDEM02,
    REP01 = CNGREP01,
    REP02 = CNGREP02,
    DISTRICT = CDDIST
          )|>
    mutate(DEM01 = as.double(str_extract_all(DEM01, "\\d+")),
           DEM02 = as.double(str_extract_all(DEM02, "\\d+")),
           REP01 = as.double(str_extract_all(REP01, "\\d+")),
           REP02 = as.double(str_extract_all(REP02, "\\d+")))|>
    mutate(TOTVOTE = REP01+DEM01+REP02+DEM02) |>
    filter(DISTRICT != 0)
```

Cleaning the redistribution district files.

For that we use the *block* unit, and the *conversion* files from the California State Assembly.

```
district_conv <- read.csv("/Users/sachabondeville/Documents/stat133/Projects/
Project-5/gerrymandering-sachabdvl/data/ab604.csv", header = FALSE)

blk <- read.csv("/Users/sachabondeville/Documents/stat133/Projects/Project-5/
gerrymandering-sachabdvl/data/state_g24_voters_by_block20.csv")

district_conv <- district_conv |>
    mutate(BLOCKID = as.character(V1))|>
    rename(NEWDISTRICT = V2)|>
    select(-1)

blk <- blk |>
    mutate(BLOCK_KEY = as.character(BLOCK20))

blk_converter <- read.csv("/Users/sachabondeville/Documents/stat133/Projects/
Project-5/gerrymandering-sachabdvl/data/state_g24_sr_blk_map.csv")

blk_converter <- blk_converter |>
    mutate(BLOCK_KEY = as.character(BLOCK_KEY))
```

Joining sequence.

We first join the new district with the block data. Then we join that with the sr_precinct data !

```
join1 <- full_join(blk, district_conv, by = join_by(BLOCK_KEY == BLOCKID))

join2 <- full_join(join1, blk_converter, by = join_by(BLOCK_KEY == BLOCK_KEY))

sr_precinct <- full_join(sr_precinct, join2, by = join_by(SRPREC_KEY ==
SRPREC_KEY))
```

Saving it back

```
write.csv(sr_precinct, "/Users/sachabondeville/Documents/stat133/Projects/
Project-5/gerrymandering-sachabdvl/data/cleaned_srprecinct.csv")
```